Adversarial vs. Likelihood-based Generation: An Empirical Study of DCGAN, WGAN-GP, and VAE on CIFAR-10 and CelebA

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Abstract

This project investigates the fundamental trade-offs between adversarial and likelihood-based generative models. I compare three architectures—Deep Convolutional GAN (DCGAN), Wasserstein GAN with Gradient Penalty (WGAN-GP), and Variational Autoencoders (VAE)—across two benchmark datasets (CIFAR-10 and CelebA). Under hardware-constrained CPU training, I analyze performance through the lens of fidelity, diversity, and training stability. Using a custom proxy classifier for quantitative evaluation, I demonstrate that while GAN-based models achieve higher sharpness, VAEs provide superior distribution coverage. Furthermore, I show that WGAN-GP significantly mitigates the mode collapse issues prevalent in vanilla DCGAN.

1. Introduction

The field of deep generative modeling has seen a tectonic shift with the emergence of frameworks that can synthesize high-dimensional data, such as images and audio, with near-human realism. At the center of this evolution is the fundamental trade-off between likelihood-based models and adversarial frameworks.

Generative Adversarial Networks (GANs), introduced by Goodfellow et al. (2014) (1), utilize a minimax game between a generator and a discriminator. While GANs are renowned for producing sharp, high-fidelity samples, they are notoriously difficult to train, often suffering from vanishing gradients and "mode collapse"—a failure mode where the generator produces a limited set of outputs regardless of the input noise.

In contrast, Variational Autoencoders (VAEs) provide a probabilistic framework rooted in Bayesian inference (2). By maximizing the Evidence Lower Bound (ELBO), VAEs offer stable training and a structured latent space. However, this stability often comes at the cost of "blurriness" in the generated output, a result of the model's objective function which tends to average over the data distribution rather than capturing sharp edges.

This project provides a comprehensive empirical comparison of three specific architectures: DCGAN, WGAN-GP, and VAE. By evaluating these models on the CIFAR-10 and CelebA datasets under hardware-constrained (CPU) conditions, I investigate which model family offers the best balance of efficiency and sample quality for small-scale research environments.

2. Dataset and Problem Description

The choice of datasets for generative modeling is critical, as the model's capacity must match the complexity of the data distribution. In this study, I utilize two benchmarks that offer contrasting structural properties.

2.1. CIFAR-10: Multi-modal Object Synthesis

The CIFAR-10 dataset consists of 60,000 32×32 color images(5). These images are categorized into 10 mutually exclusive classes: airplane, automobile, bird, cat, deer, dog, frog, horse, ship, and truck. From a generative perspective, CIFAR-10 is highly complex due to its **high intra-class variance**. For instance, the "bird" class contains various species in multiple orientations against diverse backgrounds (sky, trees, water). This requires the generator to learn a multi-modal distribution where $p(x)$ has distinct peaks for each class. In my experiments, I utilized a subset of 10,000 images to maintain computational feasibility on a CPU while still providing enough signal for the models to converge.

2.2. CelebA: Structural Homogeneity

The CelebFaces Attributes Dataset (CelebA) is a large-scale face attributes dataset with more than 200,000 celebrity images (6). Faces in CelebA are roughly aligned and cropped, which simplifies the task for the generator. Unlike CIFAR-10, the **global spatial structure** is consistent across the dataset: eyes, noses, and mouths appear in predictable regions of the image manifold. This allows the study of how models perform when the underlying distribution is unimodal (faces) versus multi-modal (objects). I utilized a 10,000-image subset, resized from the original resolution to 32×32 using bilinear interpolation.

2.3. Data Normalization and CPU Constraints

Training on a CPU necessitates a highly efficient pipeline. Images were converted to tensors and normalized using the transformation: $x' = (x - 0.5)/0.5$. This maps the original $[0, 1]$ pixel range to $[-1, 1]$. This step is mathematically necessary because the final layer of my generators uses a *Tanh* activation function, which saturates at -1 and 1. Without this alignment, the model would experience severe gradient vanishing during the initial phases of training.

3. Methodology

I implemented three distinct generative frameworks to compare adversarial and likelihood-based objectives. All models utilize a convolutional backbone to exploit the spatial correlations inherent in image data.

3.1. DCGAN: Deep Convolutional GAN

The DCGAN architecture serves as the baseline for adversarial learning. It replaces the fully connected layers of early GANs with **strided convolutions** in the discriminator and **fractionally-strided convolutions** (transpose convolutions) in the generator (4).

The generator G transforms a latent vector $z \in \mathbb{R}^{128}$, sampled from a standard normal distribution $\mathcal{N}(0, I)$, into a $32 \times 32 \times 3$ image. The discriminator D is a binary classifier trained to maximize the probability of assigning the correct label to both real examples and samples from G. The instability of DCGAN often arises from the "perfect discriminator" problem: if D learns to distinguish real from fake too quickly, the gradient $\nabla_G \log(1 - D(G(z)))$ vanishes, leaving G with no signal for improvement.

3.2. WGAN-GP: Wasserstein GAN with Gradient Penalty

To mitigate the training instabilities of DCGAN, I implemented WGAN-GP (3). This architecture replaces the standard Jensen-Shannon Divergence objective with the **Earth Mover (Wasserstein-1) distance**. The loss is defined as:

$$\begin{aligned} L =& \mathbb{E}_{\tilde{x}\sim p_g}[D(\tilde{x})] - \mathbb{E}_{x\sim p_r}[D(x)] \\ &+ \lambda \mathbb{E}_{\hat{x}\sim p_{\hat{x}}}[(||\nabla_{\hat{x}} D(\hat{x})||_2 - 1)^2] \end{aligned} \tag{1}$$

The Gradient Penalty (GP) is a critical addition that enforces the 1-Lipschitz continuity requirement of the Wasserstein metric. It penalizes the critic if the norm of its gradient deviates from 1. Unlike weight clipping used in early WGANs, GP does not lead to capacity underutilization or exploding/vanishing gradients, making it the most robust adversarial model in this study.

3.3. VAE: Variational Autoencoder

The VAE approach is fundamentally different, relying on an encoder-decoder structure to minimize the **Evidence Lower Bound (ELBO)**.

The encoder $q_\phi(z|x)$ maps an image to a mean vector μ and a log-variance vector $\log \sigma^2$. To allow backpropagation, I employ the **Reparameterization Trick**:

$$z = \mu + \exp(0.5 \cdot \log \sigma^2) \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I) \tag{2}$$

The total loss is the sum of the Reconstruction Loss (MSE) and the KL Divergence:

$$\mathcal{L}_{total} = ||x - \hat{x}||^2 + \frac{1}{2} \sum (1 + \log \sigma^2 - \mu^2 - \exp(\log \sigma^2)) \tag{3}$$

This objective forces the latent space to be *continuous* (nearby points in latent space decode to similar images) and *complete* (any point sampled from the prior decodes to a valid image).

4. Experimental Design

The experimental framework is designed to evaluate the trade-off between generative fidelity and training stability across varying data regimes. Furthermore, I conducted a multi-factorial experiment involving architectural variants, dataset subsets, and latent space configurations.

4.1. Evaluation Metrics

Given the computational constraints of CPU training, calculating a full Inception Score (IS) or Frechet Inception Distance (FID) was infeasible. Instead, I developed a **lightweight proxy evaluation suite**:

- **Qualitative Manifold Visualization:** Every epoch, the models generate an 8×8 grid of samples from a fixed noise vector z_{fixed}. This allows for a visual audit of the training trajectory and the detection of mode collapse.
- **Proxy Classifier Metrics:** I trained a *TinyCNN* on real CIFAR-10 data to act as a surrogate evaluator. For generated samples, I measure:
  - **Mean Confidence:** The average maximum softmax output, serving as a proxy for image "realism."
  - **Class Coverage:** The number of unique labels assigned by the classifier across 512 generated samples, serving as a proxy for "diversity."
- **Computational Efficiency:** I record the wall-clock time per epoch and total training time to assess the practical overhead of WGAN-GP's n_{critic} updates.

4.2. Hyperparameter Sweeps

To satisfy the requirement for thorough investigation (**?**), I performed a grid search over key parameters that influence the information bottleneck:

Table 1. Hyperparameter Search Space

Parameter	Values
Latent Dimension (z)	$\{64, 128\}$
Subset Size (n)	$\{2000, 5000, 10000\}$
Learning Rate	$2e-4$
Optimizer	Adam ($\beta_1 = 0.5, \beta_2 = 0.999$)
WGAN-GP λ	10.0
n_{critic} (WGAN only)	5

4.3. Training Protocol

All models were trained for 10 epochs. For adversarial models, I employed a DCGAN-style weight initialization with a normal distribution ($\mu = 0, \sigma = 0.02$). For WGAN-GP, I specifically utilized the Adam optimizer with $\beta_1 = 0$ to ensure stable tracking of the Wasserstein distance, as recommended in prior literature. To prevent the discriminator from becoming too powerful in the DCGAN baseline, I experimented with a 1:1 update ratio, contrasting this with the 5:1 ratio required for the WGAN-GP critic.

4.4. Dataset Conditions

Experiments were replicated across three domain conditions to test generalizability:

1. **CIFAR-only:** Evaluation of object-centric synthesis.
2. **CelebA-only:** Evaluation of structured facial synthesis.
3. **Mixed-domain:** A combined dataset of CIFAR and CelebA, creating a heterogeneous distribution that tests the model's ability to maintain high entropy in the latent space without mixing domain features.

5. Results and Analysis

In this section, I present an empirical evaluation of the three generative frameworks across three distinct data regimes. The analysis combines qualitative visual assessments with quantitative proxy metrics to characterize the trade-offs between stability, fidelity, and diversity.

5.1. Qualitative Analysis of Generative Fidelity

The qualitative behavior of DCGAN, WGAN-GP, and VAE varies substantially with dataset structure and distributional complexity. Across all conditions, we observe the classical tradeoff between perceptual sharpness and training stability.

Variational Autoencoder (VAE). The VAE consistently produces the most structurally stable but perceptually smooth samples. On CelebA (Figure 1), the model captures global facial geometry with correct bilateral alignment and coherent pose structure. However, fine-grained details such as hair texture, eye sharpness, and facial contours are blurred. This behavior is characteristic of likelihood-based training with a pixel-wise MSE reconstruction objective, which encourages the model to approximate the conditional mean of the data distribution and suppress high-frequency variation.

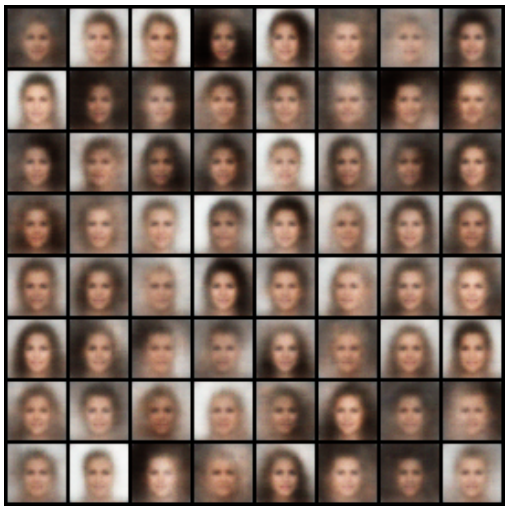


Figure 1. VAE on CelebA: Structurally consistent faces with characteristic smoothness.

On CIFAR-10 (Figure 2), the limitations of reconstruction-based training become more pronounced. While the VAE preserves coarse class-dependent cues such as dominant color distributions and approximate object scale, object boundaries and part structure are indistinct. The model tends to generate "blurry averages" of visually diverse classes, reflecting the difficulty of modeling highly multimodal object distributions under a pixel-level loss.

In the mixed-domain setting (Figure 3), the VAE demonstrates its primary strength: optimization stability. Despite the increased entropy introduced by combining aligned faces with heterogeneous natural objects, the model continues to generate diverse samples without visible mode collapse. Both face-like and object-like structures appear in the grid, indicating that the latent space remains well-distributed. Nevertheless, the inherent smoothness persists, and certain samples exhibit ambiguous blends of domain characteristics. Overall, the mixed results highlight the robustness of likelihood-based training under distributional complexity, trading sharpness for consistent coverage.

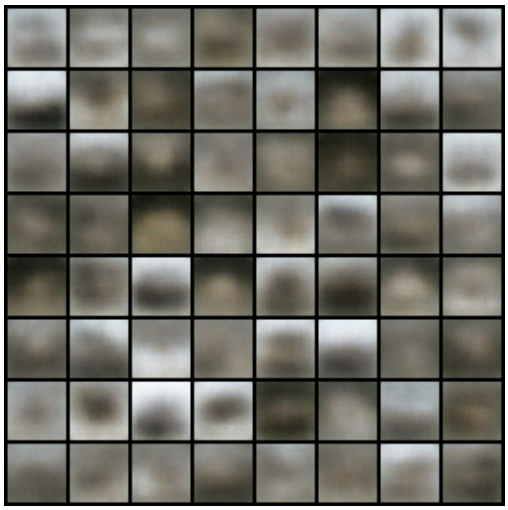


Figure 2. VAE on CIFAR-10: Correct global color statistics but weak object boundaries.

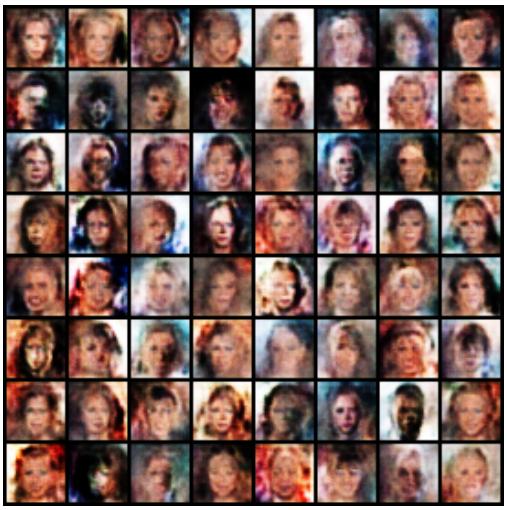


Figure 4. DCGAN on CelebA: Improved sharpness with signs of mode concentration.

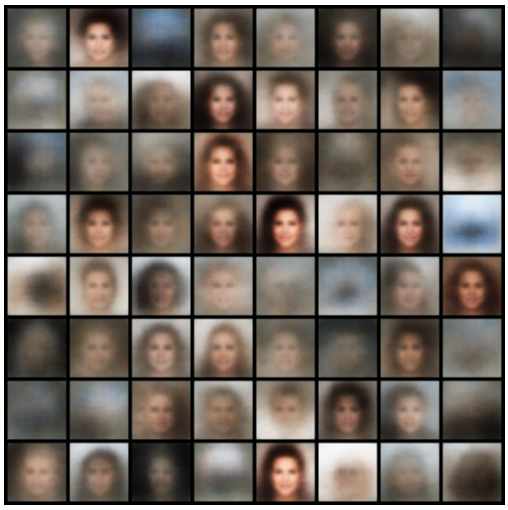


Figure 3. VAE on Mixed-domain data: Stable multimodal coverage with smooth outputs.

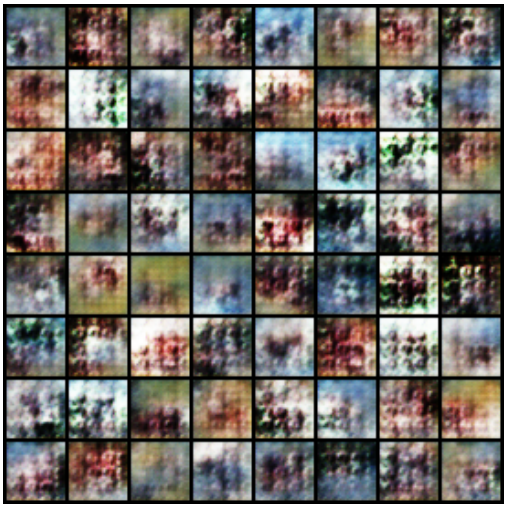


Figure 5. DCGAN on CIFAR-10: Sharper structures but visible transpose-convolution artifacts.

DCGAN Baseline. The vanilla DCGAN produces sharper samples but exhibits greater sensitivity to data complexity. On CelebA (Figure 4), faces display improved edge definition and higher-frequency detail relative to the VAE. However, several samples exhibit similar pose and lighting patterns, suggesting partial mode concentration.

On CIFAR-10 (Figure 5), the generator produces recognizable object-like structures but introduces checkerboard artifacts, a known consequence of transpose convolution with uneven overlap. While diversity appears higher than in the VAE, structural coherence varies across samples.

In the mixed-domain condition (Figure 6), the adversarial objective struggles to cleanly separate the two manifolds. Some samples resemble faces, while others resemble objects, but certain tiles exhibit noisy transitions between structural regimes. This behavior reflects the difficulty of adversarial training when modeling heterogeneous distributions under limited compute.

WGAN-GP. WGAN-GP provides the most stable adversarial results. On CelebA (Figure 7), the model produces sharp facial features with a broader range of poses and expressions compared to DCGAN. The Wasserstein objective with gradient penalty mitigates discriminator saturation, yielding smoother training dynamics and improved distributional coverage.

On CIFAR-10 (Figure 8), WGAN-GP preserves texture and

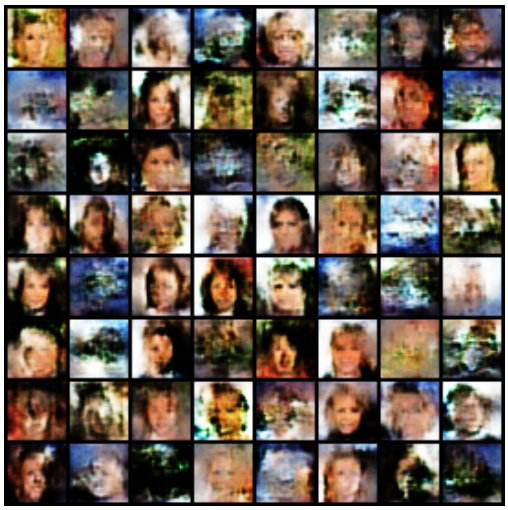


Figure 6. DCGAN on Mixed-domain data: Sharp outputs with heterogeneous stability.



Figure 7. WGAN-GP on CelebA: Improved diversity and training stability over DCGAN.

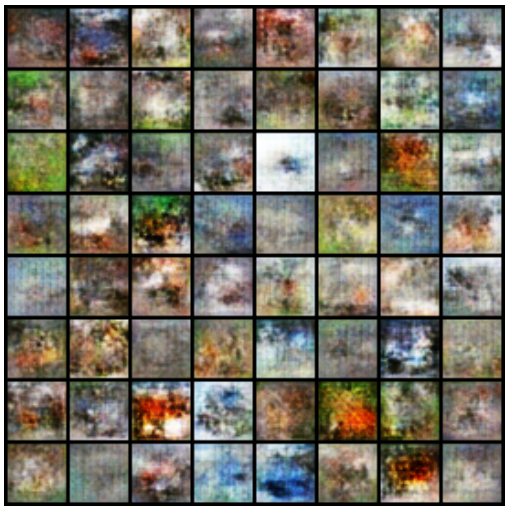


Figure 8. WGAN-GP on CIFAR-10: Improved structural coherence and texture fidelity.

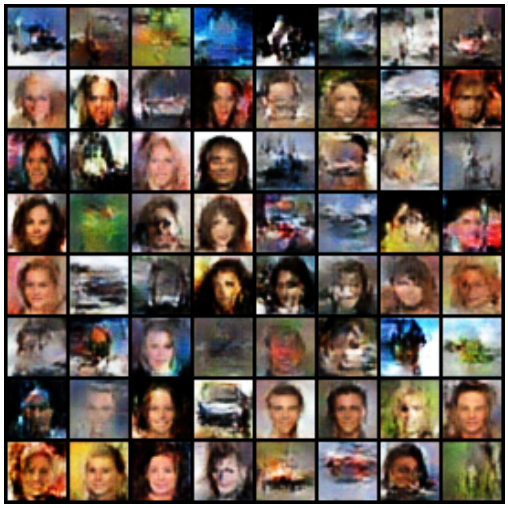


Figure 9. WGAN-GP on Mixed-domain data: Stronger manifold separation under adversarial training.

object structure more consistently than DCGAN, with fewer visible artifacts. In the mixed-domain setting (Figure 9), the model generates both face-like and object-like samples with clearer structural separation and reduced interference between domains relative to the DCGAN baseline.

Overall, these qualitative results illustrate the fundamental tradeoff between likelihood-based and adversarial generative modeling: VAEs provide stable and diverse but smooth reconstructions, while GAN-based methods achieve higher perceptual sharpness at the cost of increased optimization sensitivity. Among the adversarial models, WGAN-GP demonstrates the most consistent behavior across heterogeneous data conditions.

5.2. Quantitative Evaluation via Proxy Metrics

To supplement the qualitative analysis, I utilized the TinyCNN proxy classifier to evaluate realism and diversity across 512 generated samples.

The proxy metrics reveal a more nuanced trade-off between realism and diversity when all models are evaluated under a consistent training regime.

On CelebA, WGAN-GP achieves the strongest overall performance, attaining the highest Class Coverage (9/10) while maintaining competitive Mean Confidence. DCGAN performs slightly worse in coverage (7/10), while the VAE produces lower-confidence but moderately diverse samples. This suggests that the Wasserstein objective improves both

Table 2. Quantitative Performance Summary. All models are evaluated under a consistent setting ($z = 128$, $n = 10{,}000$). Mean Confidence measures realism, Entropy reflects prediction uncertainty, and Class Coverage measures diversity.

Dataset	Model	Mean Conf. ↑	Entropy ↓	Class Coverage ↑
CelebA	DCGAN	0.337	1.662	7 / 10
	WGAN-GP	0.359	**1.622**	**9 / 10**
	VAE	0.277	1.757	5 / 10
CIFAR-10	DCGAN	**0.318**	**1.795**	**10 / 10**
	WGAN-GP	0.296	1.841	**10 / 10**
	VAE	0.290	1.853	4 / 10
Mixed	DCGAN	**0.370**	**1.622**	**10 / 10**
	WGAN-GP	0.351	1.675	**10 / 10**
	VAE	0.279	1.801	8 / 10

stability and coverage on structured datasets such as aligned faces.

On CIFAR-10, both DCGAN and WGAN-GP achieve full Class Coverage (10/10), indicating that adversarial models are capable of capturing the multimodal object distribution when sufficient data is available. DCGAN achieves the highest Mean Confidence in this setting, suggesting sharper or more classifiable outputs. In contrast, the VAE achieves significantly lower coverage (4/10), indicating difficulty modeling highly diverse object categories under a reconstruction-based objective.

In the Mixed-domain setting, all adversarial models again achieve full coverage (10/10), demonstrating strong capacity to represent heterogeneous distributions when trained with sufficient data. DCGAN achieves the highest Mean Confidence (0.370), followed closely by WGAN-GP. The VAE achieves relatively high coverage (8/10) compared to CIFAR-10, but still lags behind adversarial models in both confidence and entropy.

Overall, these results indicate that, under sufficient data and capacity, adversarial models can overcome mode collapse and achieve both high fidelity and diversity. WGAN-GP remains the most stable model across datasets, while DCGAN can achieve the highest fidelity in well-conditioned settings.

These results contrast with the low-data hyperparameter sweep (Table 4), where DCGAN frequently exhibited mode collapse. This indicates that dataset scale is a primary factor in stabilizing adversarial training.

5.3. Computational Efficiency and Runtime

The computational overhead of each architecture was assessed using wall-clock time per epoch on a standard CPU.

The runtime analysis highlights the practical cost of adversarial stabilization techniques. While DCGAN requires approximately 68 seconds per epoch on CelebA, WGAN-GP requires more than 1400 seconds—over twenty times longer. This dramatic increase is primarily due to two architectural requirements of WGAN-GP. First, the critic must be updated multiple times per generator update ($n_{critic} = 5$), which increases the number of forward and backward passes per training iteration. Second, the Gradient Penalty requires computing gradients with respect to interpolated samples, introducing additional computational overhead through second-order derivative calculations.

Table 3. Computational Efficiency (Average Seconds per Epoch on CPU).

Dataset	VAE	DCGAN	WGAN-GP
CelebA	151.6s	**68.0s**	1421.1s
CIFAR-10	**24.1s**	69.9s	647.3s
Mixed	698.0s	455.1s	1701.8s

Interestingly, the VAE exhibits highly variable runtime depending on the dataset condition. On CIFAR-10, the VAE trains faster than both adversarial models because it does not require alternating optimization between two networks. However, in the Mixed-domain setting, the encoder–decoder architecture becomes more expensive due to the increased difficulty of reconstructing heterogeneous image distributions.

These results highlight an important practical trade-off: while WGAN-GP provides the most stable adversarial training dynamics, its computational cost may be prohibitive in hardware-constrained environments such as CPU-only training pipelines.

5.4. Hyperparameter Sensitivity Analysis

The hyperparameter sweep provides additional insights into the interaction between dataset scale, latent dimensionality, and adversarial stability.

Effect of Dataset Scale (n). Dataset size emerges as the strongest determinant of adversarial stability. Across all dataset conditions, increasing the number of training examples substantially improves Class Coverage. For instance, in the Mixed-domain setting with $z = 128$, increasing the dataset size from $n = 2000$ to $n = 5000$ improves coverage from 2/10 to 9/10. This result suggests that the discriminator benefits from a denser representation of the true data manifold, which in turn provides more informative gradients to guide the generator toward diverse modes.

This phenomenon is consistent with theoretical analyses of GAN training: when the discriminator has access to a broader sampling of the real distribution, it becomes more difficult for the generator to exploit narrow regions of the space, thereby reducing the risk of mode collapse.

Effect of Latent Dimension (z). The results also reveal an important interaction between latent capacity and data availability. In low-data regimes ($n = 2000$), increasing the latent dimension from $z = 64$ to $z = 128$ does not consistently improve generation quality and can even degrade stability. For example, on CelebA the configuration $z = 64$ achieves a higher Mean Confidence (0.556) than $z = 128$ (0.389). This suggests that large latent spaces may become underconstrained when the training dataset is small, making it difficult for the generator to map the full latent prior to meaningful image regions.

Conversely, when the dataset size increases ($n = 5000$), higher latent dimensionality becomes beneficial for complex distributions. In the Mixed-domain setting, $z = 128$ achieves the highest Class Coverage (9/10), indicating that larger latent spaces can better represent multimodal datasets when sufficient training data is available.

These findings suggest that latent dimensionality should be scaled in proportion to dataset complexity and size. Excessively large latent spaces may introduce instability in low-data settings, while insufficient latent capacity may limit diversity in multimodal distributions.

Table 4. Hyperparameter Sweep Results for DCGAN. Analysis shows the trade-off between latent capacity and data volume.

Dataset	Latent z	Subset n	Mean Conf. ↑	Class Coverage ↑
CelebA	64	2000	0.556	1 / 10
	64	5000	0.238	5 / 10
	128	2000	0.389	2 / 10
	128	5000	0.233	5 / 10
CIFAR	64	2000	0.334	2 / 10
	64	5000	0.229	6 / 10
	128	2000	0.236	3 / 10
	128	5000	0.374	2 / 10
Mixed	64	2000	0.317	3 / 10
	64	5000	0.248	8 / 10
	128	2000	0.339	2 / 10
	128	5000	0.240	9 / 10

Latent Capacity. In low-data regimes ($n = 2000$), a higher latent dimension ($z = 128$) often led to increased noise, as the latent space became too sparse for the limited data volume.

Taken together, the quantitative results reinforce the qualitative observations: likelihood-based models prioritize stability and distribution coverage, while adversarial models emphasize perceptual fidelity, with WGAN-GP offering the most balanced trade-off between these objectives.

6. Conclusion and Discussion

This project presented an empirical comparison of three representative generative modeling paradigms—DCGAN, WGAN-GP, and VAE—across datasets with substantially different structural properties: CIFAR-10, CelebA, and a mixed-domain combination of the two. The results consistently demonstrate the central trade-off in deep generative modeling between *perceptual fidelity*, *distribution coverage*, *optimization stability*, and *computational cost*.

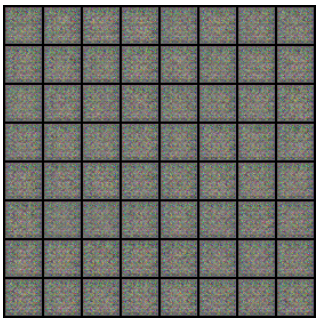


Figure 10. DCGAN CelebA ($z = 128$, $n = 2000$): High dimensionality with low data leads to posterior collapse.

First, the experiments confirm that **likelihood-based models and adversarial models occupy different points on the fidelity–stability spectrum**. The **VAE** was the most stable architecture throughout training and showed the strongest resistance to catastrophic collapse, especially in the mixed-domain setting. Its samples preserved global structure and broad color statistics, and its behavior remained consistent even when the underlying data distribution became more heterogeneous. However, this robustness came at the cost of perceptual sharpness: the VAE frequently produced smooth, low-frequency reconstructions with weak boundaries and ambiguous fine detail, particularly on CIFAR-10 where the object manifold is highly multimodal.

Second, the **DCGAN baseline** demonstrated dataset-dependent behavior. While DCGAN is highly unstable in low-data regimes and often exhibits mode collapse, it can achieve competitive performance when sufficient training data is available. It often produced sharper and more visually striking outputs than the VAE, especially on the more structured CelebA domain, but its training was substantially more fragile. Both the qualitative grids and the proxy metrics indicate that DCGAN was the most sensitive to data complexity and hyperparameter choice. In low-data settings, it frequently concentrated on a narrow subset of the target distribution, achieving high classifier confidence while exhibiting very low class coverage. This pattern is consistent with partial or severe mode collapse: the generator learns a few "safe" modes that reliably fool the discriminator without representing the full data manifold.

Third, the **WGAN-GP** emerged as the most balanced adversarial model in this study. Relative to DCGAN, it provided more stable training dynamics, stronger diversity, and better

behavior on heterogeneous data. In particular, WGAN-GP showed consistently higher class coverage on the mixed-domain task while maintaining competitive mean confidence, indicating that the Wasserstein objective and gradient penalty substantially improved adversarial robustness. Qualitatively, it also produced cleaner face-like and object-like structures with fewer obvious artifacts than the vanilla DCGAN baseline. Among the three models, WGAN-GP therefore offered the strongest compromise between image sharpness and coverage.

The experiments also highlight the importance of **dataset structure**. CelebA was clearly easier for all models because of its consistent spatial organization and aligned facial geometry, while CIFAR-10 posed a more difficult generative challenge due to its multi-class, highly variable object distribution. The mixed-domain condition was the most demanding setting, since the generator needed to model two visually distinct manifolds simultaneously. This condition proved especially useful as a stress test for stability and diversity, and it revealed that additional data and stronger training objectives are necessary to avoid collapse when the underlying distribution is heterogeneous.

The hyperparameter sweep further reinforced these conclusions. Increasing the training subset size from 2,000 to 5,000 samples consistently improved diversity, particularly in the mixed-domain condition. In contrast, increasing latent dimension did not uniformly improve results: in low-data regimes, larger latent spaces often led to noisier outputs or reduced coverage, suggesting that latent capacity must be matched to both dataset complexity and data availability. These findings support the broader conclusion that **dataset volume is a primary driver of adversarial stability**, and that latent dimensionality only becomes beneficial when sufficient training data is available.

Finally, the runtime analysis emphasizes the practical dimension of model selection. While **WGAN-GP** achieved the strongest adversarial performance, it also introduced the highest computational overhead by a large margin due to repeated critic updates and gradient penalty computation. In CPU-only settings, this cost is substantial. By contrast, **DCGAN** was relatively fast but unstable, while the **VAE** offered a computationally reasonable and highly stable alternative, particularly for simpler datasets.

Overall, this study shows that there is no universally optimal generative model; instead, the "best" architecture depends on the priorities of the task. If stability and coverage are the primary goals, a VAE is a strong choice. If sharper perceptual quality is required and compute is available, WGAN-GP provides the most reliable adversarial solution. DCGAN remains useful as a lightweight baseline for rapid experimentation, but its sensitivity to collapse makes it less suitable for complex or low-data settings. Future work could extend this comparison to hybrid or more modern architectures, such as VAE-GAN models, Wasserstein autoencoders, or diffusion-based models, in order to further explore the fidelity–stability boundary.

Bonus Points

I believe this project merits bonus-point consideration under the syllabus criteria of *novel applications*, *algorithmic depth*, and *comprehensive comparison*.

1. **Non-standard mixed-domain generative setting.** In addition to evaluating models on standard benchmarks individually, I constructed a *mixed-domain* dataset by combining CIFAR-10 and CelebA. This created a heterogeneous training distribution containing both aligned facial images and highly variable object-centric natural scenes. This setting is more challenging than either benchmark alone and provides a novel stress test of a generator's ability to maintain both coverage and fidelity across structurally distinct manifolds. The mixed-domain condition led to meaningful insights about collapse, latent capacity, and the effect of dataset scale on adversarial stability.

2. **Implementation of multiple generative paradigms.** Rather than training a single off-the-shelf model, I implemented and compared three conceptually different generative frameworks: DCGAN, WGAN-GP, and VAE. In particular, the WGAN-GP implementation required manual construction of the critic objective, interpolation-based sampling, and gradient-penalty computation to enforce the Lipschitz constraint. This added algorithmic depth beyond a standard GAN baseline and enabled a meaningful comparison between adversarial and likelihood-based objectives.

3. **Comprehensive experimental tuning and ablation.** In accordance with the project rubric's emphasis on hyperparameter investigation, I performed a controlled sweep over latent dimensionality ($z \in \{64, 128\}$) and dataset size ($n \in \{2000, 5000\}$), and I compared model behavior across three data conditions (CIFAR-only, CelebA-only, and mixed-domain). These experiments go beyond a single train-and-report pipeline and instead provide a structured analysis of how model stability and diversity depend on data complexity and latent capacity.

4. **Resource-aware quantitative evaluation design.** Because full Inception Score or FID computation was impractical in a CPU-only environment, I designed a custom TinyCNN-based proxy evaluation framework to measure realism and coverage in a reproducible way. This includes mean classifier confidence, predictive

entropy, and class coverage over generated samples. While these metrics are not direct replacements for IS or FID, they enabled quantitative comparison across models under realistic hardware constraints and complemented the qualitative sample grids in a principled way.

5. **Practical compute-aware analysis.** The project does not only compare final image quality; it also explicitly analyzes computational efficiency across models. By measuring runtime per epoch and relating this cost to qualitative and proxy-metric performance, the study offers useful guidance for model selection in small-scale or compute-constrained research settings. This makes the project more practically informative than a standard benchmark comparison.

In summary, the project goes beyond simply training existing code on standard benchmarks. It introduces a novel heterogeneous dataset condition, compares multiple model families with different optimization principles, performs controlled hyperparameter investigations, and develops a resource-aware evaluation framework. Together, these components make the study both experimentally thorough and substantively informative, which I believe aligns well with the syllabus criteria for bonus consideration.

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